

# Writing Inequalities from Verbal Constraints

*Turning a comparison phrase into a symbol, building a one-variable inequality from a story, and solving it and reading the answer back into the situation*

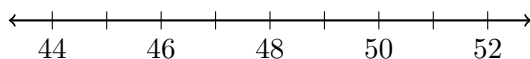
The four phrases that decide the symbol.

| Phrase                          | Symbol | Is the boundary allowed?            |
|---------------------------------|--------|-------------------------------------|
| more than, over                 | $>$    | no — open circle on the number line |
| less than, under                | $<$    | no — open circle                    |
| at least, no less than, minimum | $\geq$ | yes — closed circle                 |
| at most, no more than, maximum  | $\leq$ | yes — closed circle                 |

## Directions.

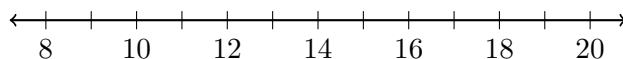
- Say in words what the variable counts or measures before you write anything — that sentence is part of the answer.
- Write the inequality first, then solve it. Graph the solution on the number line under the problem, using an open or closed circle to match your symbol.
- If a step multiplies or divides both sides by a negative number, the inequality symbol flips.

1. A ride at the fair requires riders to be *at least* 48 inches tall. Let  $h$  be a rider's height in inches. Write an inequality for the heights that may ride, and graph it.



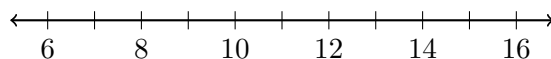
Answer: \_\_\_\_\_

- 2.** A school van can carry *no more than* 14 passengers. Let  $p$  be the number of passengers on a trip. Write an inequality for the trips that are allowed, and graph it.



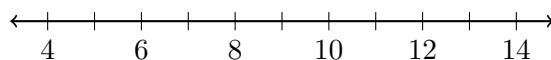
Answer: \_\_\_\_\_

- 3.** Maya has \$60 to spend at the fair. Entry costs \$16 and each snack costs \$4. Let  $s$  be the number of snacks she buys. Write an inequality saying she does not overspend, then solve it and graph the solution.



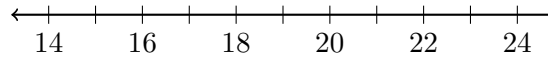
Answer: \_\_\_\_\_

- 4.** A gym charges a \$30 joining fee once, then \$12 for each month of membership. Deshawn can spend *at most* \$150 in total. Let  $m$  be the number of months. Write an inequality, solve it, and graph the solution.



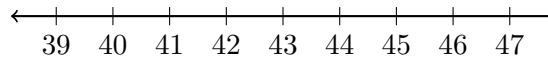
Answer: \_\_\_\_\_

- 5.** A club needs to raise *at least* \$500. They already have \$180 and sell wreaths for \$16 each. Let  $w$  be the number of wreaths sold. Write an inequality for reaching the goal, solve it, and graph the solution.



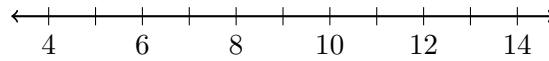
Answer: \_\_\_\_\_

- 6.** A freight elevator may carry *no more than* 2100 pounds. The operator weighs 165 pounds and each box weighs 45 pounds. Let  $b$  be the number of boxes loaded. Write an inequality, solve it, graph the solution, and then say how many boxes the operator may actually load.



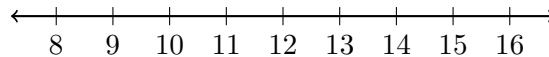
Answer: \_\_\_\_\_

**7.** A drone starts a descent from a height of 240 metres and falls 15 metres every second. Let  $t$  be the number of seconds since the descent began. Write an inequality for the times when the drone is *above* 90 metres, solve it, and graph the solution. Watch what happens to the symbol when you divide.



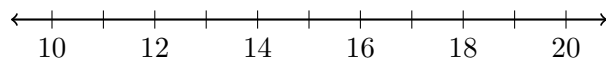
Answer: \_\_\_\_\_

**8.** Jonah is told “a number  $n$  is no more than 12” and writes  $n < 12$ . His answer is not quite right. Explain in one sentence what is wrong with his symbol, write the correct inequality, and graph it.



Answer: \_\_\_\_\_

**9.** A rectangle has a width of 8 centimetres and a length of  $l$  centimetres. Its perimeter must be *at most* 46 centimetres. Write an inequality for the length, solve it, and graph the solution.



Answer: \_\_\_\_\_

**10.** Now work backwards. Invent a real situation that the inequality  $25 + 8n \geq 65$  could describe. Say clearly what  $n$  counts and what the two numbers mean in your story, then explain what the solution  $n \geq 5$  tells somebody in that situation. Graph  $n \geq 5$  on the line below.

